

Math 571 - Homework 1

Problem 1.1 (R:1:2*). Show that for any positive integer n , if n is not a perfect square, then $\sqrt{n}$ is irrational.

Problem 1.2 (R:1:4*). Let E be a non-empty subset of an ordered set $(S, <)$; suppose that α is a lower bound for E in S and β is an upper bound for E in S . Show that $\alpha \leq \beta$. Can $\alpha = \beta$? What happens if $E = \emptyset$?

Problem 1.3 (R:1:5). Let A be a non-empty set of real numbers bounded below. Let $-A = \{-a \mid a \in A\}$. Show that

$$\inf(A) = -\sup(-A)$$

The next two problems, R:1:6 and R:1:7, show that the exponentiation map from $\mathbb{Q}$ to $\mathbb{R}$ given by $r \mapsto b^r$ for $b > 1$ can be uniquely extended to $x \mapsto b^x$ for $x \in \mathbb{R}$ and that for $b > 1$ and $y \in (0, \infty)$ we can find x so that $b^x = y$, that is, $\log_b : (0, \infty) \rightarrow \mathbb{R}$ is well-defined.

Problem 1.4 (R:1:6). Fix $b > 1$.

(a) If n, m, p, q are integers, $n, q > 0$, and $r = m/n = p/q$, prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Because of this defining b^r and $b^{m/n}$ for any $m/n = r$ provided $n > 0$ is well-defined.

(b) Prove that $b^{r+s} = b^r b^s$ if r and s are rational.

(c) If $x \in \mathbb{R}$, define $B(x) = \{b^t \mid t \in \mathbb{Q} \wedge t \leq x\}$. Prove that

$$b^r = \sup(B(r))$$

whenever r is rational.

Explain why it makes sense to define

$$b^x = \sup(B(x))$$

for every real x .

(d) Prove that $b^{x+y} = b^x b^y$ for every real x and y .

Problem 1.5 (R:1:7). Fix $b > 1$, $y > 1$, and show that there is a unique x so that $b^x = y$. This defines $\log_b(y) = x$.

- (a) For any positive integer n , $b^n - 1 \geq n(b - 1)$.
- (b) $b - 1 \geq n(b^{1/n} - 1)$.
- (c) If $t > 1$ and $n(t - 1) > (b - 1)$, then $b^{1/n} < t$.
- (d) If w is such that $b^w < y$, then $b^{w+1/n} < y$ for large enough n . To see this, apply (c) to $t = y \cdot b^{-w}$.
- (e) If $b^w > y$, then $b^{w-1/n} > y$ for large enough n .
- (f) Let $A(y) = \{w \mid b^w < y\}$ and $x = \sup(A(y))$. Show that $y = b^x$.
- (g) Show that x is unique.

Problem 1.6 (R:1:8). Show that $\mathbb{C}$ can not be made into an ordered field.

The following two problems are also paired. R:1:14 is really just a special instance of R:1:17*, which differs from Rudin by including the case $\mathbf{x}, \mathbf{y} \in \mathbb{C}^k$

Problem 1.7 (R:1:14). Show that for $w, z \in \mathbb{C}$

$$|w + z|^2 + |w - z|^2 = 2|w|^2 + 2|z|^2.$$

Use this to compute $|1 + z|^2 + |1 - z|^2$ given that $|z| = 1$.

Problem 1.8 (R:1:17*). Show that for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^k(\mathbb{C}^k)$,

$$\|\mathbf{x} + \mathbf{y}\|_2^2 + \|\mathbf{x} - \mathbf{y}\|_2^2 = 2\|\mathbf{x}\|_2^2 + 2\|\mathbf{y}\|_2^2. \quad (\text{Parallelogram Law})$$

How does this generalize the Pythagorean theorem?